



Solar cell I-V curves and equivalent circuit (Courtesy: Agilent Technologies)

are characterised by a maximum V_{oc} at zero output-current, and a short circuit current (I_{sc}) at zero output-voltage.

Testing of the solar cell is required for research, quality assurance and production. PV cell testing is an important part of PV semiconductor design and fabrication, and includes a variety of tests that help ensure the quality and efficiency of the product.

Although the measurement accuracies, speeds and other parameters may differ across different levels, there are a number of key parameters that are typically measured in any testing environment. Two simple tests can be performed to obtain all of the necessary data for solar cell I-V characterisation—the forward-bias (illuminated) test and the reverse-bias (dark) test.

Parameters of solar cell and module for measurement

Voltage at maximum power (V_{mp}). It is the highest voltage the panel can produce while connected to a system and

operating at peak efficiency.

Open circuit voltage (V_{oc}). It is the maximum voltage that the panel can produce when not connected to an electrical circuit or system, that is, when the current through the solar cell is zero. It is the maximum voltage available from the solar cell. V_{oc} can be measured with a meter directly contacting the panel's terminals or the ends of its built-in cables.

Short circuit current (I_{sc}). It is the current through the solar cell when the voltage across the solar cell is zero.

Maximum power current (I_{mp}). It is the maximum current available when the panel is operating at peak efficiency in a circuit.

Maximum power point (P_{max}). It is the condition under which the solar cell generates its maximum power. The current and voltage in this condition are defined as I_{max} and V_{max} , respectively.

Fill factor (FF) and the conversion efficiency (η). These are metrics used to characterise the performance of the solar cell. Fill factor is defined as the ratio of P_{max} divided by the product of V_{oc} and I_{sc} .

Conversion efficiency. It is the percentage of power converted (from absorbed light to electrical energy) and collected when a solar cell is connected

to an electrical circuit. It is the ratio of P_{max} to the product of the input light irradiance (E) and the solar cell surface area (A).

Internal resistances of a solar cell. R_s and R_{sh} are internal parasitic series and shunt resistances, respectively. During operation, the efficiency of solar cells is reduced by the dissipation of power in these resistances.

Series resistance in a solar cell has three causes: first, the movement of current through the emitter and base of the solar cell; second, the contact resistance between the metal contact and the silicon; and finally the resistance of the top and rear metal contacts. The main impact of series resistance is to reduce the fill factor. Although excessively high values may also reduce the short circuit current, R_{sh} for an ideal cell would be infinite and would not provide an alternate path for current to flow, while R_s would be zero, resulting in no further voltage drop before the load. Decreasing R_{sh} and increasing R_s will decrease the FF and P_{max} .

Capacitance measurement. This includes the measurement of parallel capacitance (C_p), carrier density (N_c) and drive-level density (N_d).

Time domain measurement. Minorities carrier lifetime (τ), surface recomb-



A typical PV system